Limits of Linguistic and Zipfian Features for ASD Detection: A Neural Network Replication on ASDBank

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Abstract **- Early, objective screening for autism spectrum disorder (ASD) remains challenging because current assessments are slow, clinician-dependent, and hard to scale. Speech transcripts offer a low-burden, privacy-preserving signal, so we ask whether simple linguistic measures and distributional cues from Zipf's law can reliably separate ASD from typical development. Prior work has reported 75–80% accuracy using linguistic features, and other studies suggest ASD speech may deviate from Zipf's law. This law describes a power-law pattern in language in which a few words are very common and most are rare. If ASD transcripts diverge from that pattern, the degree of Zipfian fit could be a diagnostic cue. We attempted to replicate and extend these claims on a balanced subset of ASDBank (Eigsti; 16 ASD, 16 TD) by training neural networks on (i) linguistic features, (ii) Zipf-based metrics, and (iii) their combination across 100 randomized train–test splits with data augmentation. Across all conditions, performance hovered around chance. Feature analysis showed ASD children used fewer conjunctions, but no other linguistic feature or Zipfian metric displayed clear group separation or predictive value. Contrary to prior reports, neither feature set, alone or combined, yielded reliable classification from short transcripts. These results underscore reproducibility limits and suggest that larger datasets and richer modalities will be required for robust, transcript-based ASD screening.**

Code and Data Availability -All code for feature extraction, Zipf analysis, and neural-network training (including scripts to reproduce the 100-split experiments and Figures 1–3) is available at: https://github.com/Krishnapabbu17/Autism_Project

I. Introduction

Autism Spectrum Disorder (ASD) is a developmental condition affecting about 1% of the global population [1], including approximately 1 in 54 children in the United States [2]. Early diagnosis and intervention can significantly improve long-term outcomes for children with ASD [3]. However, current diagnostic practices are time-consuming, labor-intensive, and subjective, often requiring extensive evaluations by clinicians and repeated observations of the child [3]. As a result, an estimated one-fourth of children with ASD go undiagnosed in early childhood [4]. There is a pressing need for more efficient, objective screening methods to identify ASD as early as possible.

In recent years, researchers have turned to machine learning (ML) and artificial intelligence to aid in ASD detection [3]. Various modalities have been explored. For example, computer vision analysis of facial images, eye-gaze tracking, and neuroimaging data – which have revealed distinctive patterns associated with ASD. Studies show that children with ASD produce facial expressions that are more ambiguous and less socially meaningful than those of typically developing peers [5]. Likewise, eye-tracking research has found that children with ASD tend to focus less on human faces and more on background objects during social interactions [6]. Even neuroimaging biomarkers have been investigated. For instance, ML models can differentiate individuals with ASD using brain MRI or EEG features [6]. While these approaches show promise, they also raise practical and ethical concerns. Collecting images of a child's face, retina, or brain over time can be invasive and raises privacy issues – especially for minors [3]. In contrast, analyzing a child's natural speech offers a rich but far less intrusive data source. Unlike images, speech can be gathered unobtrusively and anonymized (preserving content but not identity) [3]. Every child's speech patterns are unique, and language delays or anomalies are a hallmark of ASD [3]. Notably, children with ASD often exhibit unusual speech characteristics – for example, they may repeat words and phrases (echolalia) and use a

more limited or simplified vocabulary and syntax compared to their typically developing peers [3]. These traits make speech transcripts a promising resource for automated ASD screening that preserves the child's privacy.

Prior studies in computational linguistics have indeed demonstrated that analyzing child speech can help distinguish ASD from typical development. For example, Ramesh and Assaf (2021) used transcribed child speech from TalkBank's ASDBank corpus (a repository of annotated child language transcripts) [7] and extracted over 50 linguistic features (e.g. mean utterance lengths, vocabulary richness, and part-of-speech frequencies) to train various ML classifiers [3]. Their models achieved up to about 75–80% accuracy in predicting ASD vs. typical development, with logistic regression and random forests performing best [3]. Crucially, their feature analysis identified several linguistic indicators of ASD. Children with ASD tended to have shorter utterances on average and took different turn lengths in conversation, and they showed differing usage of certain parts of speech compared to typically developing children [3]. For example, metrics such as Mean Length of Utterance (MLU), the mean turn length ratio (child's utterance length relative to the adult's), and the frequency of pronouns, conjunctions, and negations in the child's speech were among the top predictors of an ASD diagnosis in their models [3]. These findings suggest that the linguistic complexity and content of a child's speech are key signals for ASD detection.

In addition to traditional linguistic features, recent research highlights a novel perspective: Zipf's law statistics of word usage. Zipf's law states that in natural language, word frequencies typically follow a power-law distribution where a few words are very common and many others are rare. Interestingly, it has been shown that typical child speech conforms to a Zipfian distribution, whereas the speech of children with ASD deviates from Zipf's law [8]. In one study, the word-frequency distributions in transcripts of neurotypical children were well-fit by a power law (indicating expected "Zipfianness"), but the transcripts of age-matched children with ASD were found to be non-Zipfian [8]. Using rigorous statistical tests (e.g. Kolmogorov–Smirnov goodness-of-fit and likelihood ratio tests), researchers confirmed that the ASD speech corpora did not follow the expected power-law distribution [8]. This divergence likely stems from the repetitive and less diverse language in ASD. For instance, a child with ASD might use a narrow set of favorite words extremely frequently, skewing the overall frequency-rank profile. Such differences in lexical distribution could serve as quantitative markers of ASD-language abnormalities [8]. De Palma et al. (2021) posit that metrics of Zipfian "fit" might thus complement traditional behavioral cues in diagnosing ASD [8]. In other words, how well a child's speech adheres to (or breaks from) Zipf's law might be an indicative signal of ASD that an automated system can detect.

Motivated by these insights, the present work proposes a multimodal neural network approach that integrates Zipfian statistical features with conventional linguistic features to improve ASD detection from child speech transcripts. We investigate whether combining these feature sets can boost classification performance compared to using linguistic features alone. In the following, we describe our methodology and evaluate the model on child speech data from the ASDBank corpus [7]. The introduction of Zipf-based features is a novel contribution aimed at capturing distributional differences in ASD speech. We hypothesize that a model leveraging both linguistic complexity measures and lexical distribution metrics will more accurately distinguish children with ASD from typically developing (TD) children. The next sections present our experimental results in support of this hypothesis.

II. METHODOLOGY

Data: We conducted our experiments on the ASDBank English Eigsti Corpus from TalkBank [7]. This dataset consists of transcribed spontaneous speech from children in a controlled play interaction setting (30-minute free play sessions). The corpus includes three groups of children ages 3–6: ASD (children diagnosed with autism spectrum disorder), DD (children with non-ASD developmental delays), and TD (typically developing children), with 16 children in each group. In our study we focused on distinguishing ASD vs. TD children, so we used only the transcripts from the ASD group (n=16) and the TD group (n=16), for a total of 32 child transcripts. Each transcript contains the first 100 utterances from the child in the play session (interacting with an adult experimenter). The transcripts are formatted in the CHAT notation and include the child's utterances and the adult's utterances. Prior to feature extraction, basic preprocessing was done on the transcripts: we removed any annotations or non-speech elements and standardized the text (ex. lowercasing). No additional text normalization was needed beyond what was already provided in the TalkBank transcripts.

Linguistic Feature Extraction: From each child's transcript, we extracted a set of linguistic features that have been identified in previous work [3] as potentially discriminative for ASD vs. TD speech. These included measures of utterance length, part-of-speech usage, and overall verbal output. For instance, we calculated the child's mean length of utterance (MLU), which is the average number of morphemes per utterance—a standard metric of linguistic development (children with ASD often have lower MLU) [3]. We also computed the mean length of turn (MLT) ratio, defined as the ratio of the child's mean utterance length to the adult partner's mean utterance length in the conversation [3]. This feature captures how the child's conversational turn-taking compares to the adult's. A markedly lower MLT ratio (meaning the child speaks in much shorter turns than the adult) might indicate less conversational engagement or reduced linguistic ability.

In addition, we quantified the child's use of certain parts of speech by measuring the percentage of words in the transcript that were pronouns, conjunctions, or negations (ex. "no", "not"). These categories were chosen because prior studies have noted atypical language patterns in ASD, such as avoidance or reversal of pronouns and a possible reduction in the use of connective words like conjunctions [3]. We normalized these counts by the total number of words (reporting them as rates per 100 words) to account for transcript length. Finally, as a simple measure of verbal output, we recorded the total number of words the child produced in the 100-utterance sample [3].

We calculated all these feature values using custom scripts to parse the CHAT transcripts. The resulting linguistic feature vector for each child consisted of six features: MLU, MLT ratio, pronoun rate (per 100 words), conjunction rate (per 100 words), negation rate (per 100 words), and total word count.

Zipfian Feature Extraction: In parallel, we extracted features characterizing the word-frequency distribution of each child's speech. For each child's transcript, we obtained the frequency count of every word the child spoke (excluding the adult's words) and analyzed whether this distribution followed a Zipfian (power-law) pattern. To quantify this, we applied the statistical approach described by Clauset et al. (2009) [9], as used in a recent study by De Palma et al. (2021) [8]. Specifically, we performed a maximum-likelihood power-law fit for the upper tail of the word frequency distribution using a power-law analysis software package [10]. From this fit, we recorded the estimated power-law exponent (α), which reflects how quickly word frequency drops off for less common words—a higher α indicates a steeper drop, potentially signaling heavier repetition of a few words. We also determined xmin, the minimum frequency rank above which the power-law model was fit (this cutoff is chosen by the fitting procedure to optimize the fit [9]).

Next, we evaluated how well each child's word frequency distribution actually conforms to Zipf's law. We calculated a goodness-of-fit p-value using a Kolmogorov–Smirnov test with bootstrap simulation [9]. A high p-value (typically > 0.1) suggests the data are consistent with a Zipfian distribution, whereas a low p-value indicates a poor fit (i.e., a significant deviation from Zipf's law). In addition, we compared the power-law fit to an alternative (log-normal) distribution by means of a likelihood ratio (LR) test [9]. This provided an LR statistic and an associated p-value; if the log-normal model fit significantly better (ex. a significant LR test favoring log-normal), it would imply that the word frequencies are not well-described by a power law, consistent with a non-Zipfian pattern in the child's speech [8].

From these analyses, we derived a set of Zipf-based features for each transcript. In particular, we used the fitted power-law exponent α, the goodness-of-fit p-value, and the significance result of the power-law versus log-normal LR test as features. (We initially considered including xmin and the raw LR statistic as well, but these proved less informative.) In summary, each child's Zipf feature vector captures the degree to which their lexical distribution adheres to Zipf's law or deviates from it.

After extracting all features, we performed data cleaning and normalization. Any instances of undefined or infinite feature values were removed or replaced with neutral values. For example, if a feature like the MLT ratio were undefined due to zero

adult utterances (which did not occur in our data) or if a child's transcript contained anomalies, that sample was excluded. We also applied an outlier filter, removing any transcript for which a feature value lay more than 3 standard deviations from the group mean; this eliminated a few extreme cases and improved robustness. Finally, all feature values were standardized (z-scored) before modeling so that features on different scales could be combined meaningfully.

Model Training: We framed ASD detection as a binary classification problem (ASD vs. TD). Given the relatively small size of our dataset (32 transcripts total, reduced to 29 after data cleaning), we adopted a data augmentation strategy to bolster the training data. In particular, we used a Bayesian generative approach to synthesize additional training examples in feature space. For each training fold, we first modeled the distribution of the features in the available training set using probabilistic inference. We assumed each feature dimension (after z-score normalization) was approximately normally distributed across the training examples. Using a Bayesian sampling method (via the PyMC library), we estimated the mean and standard deviation for each feature based on the training data. We ran Markov Chain Monte Carlo (MCMC) simulations to draw samples from the posterior distribution of these parameters, capturing the uncertainty in the feature statistics given the limited data.

Using the learned feature distributions, we then generated synthetic data points to augment the training set. In each training iteration, we sampled multiple sets of feature means and variances from the posterior and drew new feature vectors from those distributions. In total, we created 1,000 synthetic feature vectors for each training split. We randomly assigned labels to these synthetic examples to roughly preserve the class balance (approximately half labeled ASD and half TD, since the real training data were balanced).

Next, we trained a feed-forward neural network classifier on the combined dataset of real and synthetic examples. The network was a multilayer perceptron with an input layer equal to the number of features, two hidden layers of 64 and 32 neurons (with ReLU activation), a smaller dense layer of 16 neurons, and a final output neuron with sigmoid activation for binary (ASD vs. TD) prediction. We applied dropout regularization (with dropout rates of 0.3 and 0.2 in the two hidden layers) during training to mitigate overfitting. The model was optimized using the Adam optimizer on a binary cross-entropy loss, and we employed early stopping based on validation performance to prevent over-training.

To evaluate the model's performance, we used a repeated random hold-out validation procedure. We ran 100 independent trials, each time randomly splitting the children into an 80% training set and a 20% test set. For each trial, we performed the Bayesian feature modeling, synthetic data generation, and neural network training afresh, then evaluated the classifier on that trial's unseen test data. This yielded 100 independent accuracy measurements. We chose this resampling strategy (instead of a single fixed split or standard k-fold cross-validation) to make full use of the limited data and to observe how performance varies with different train–test partitions. The primary evaluation metric was classification accuracy on the held-out test set. In the next section, we present the classification results and examine which features were most informative for distinguishing ASD vs. TD speech.

III. RESULTS

Across all three feature configurations, linguistic only, Zipf only, and their combination, the classifier's performance hovered around chance with large split-to-split variability. This instability indicates weak generalization from short transcripts in a small-n setting. We summarize each condition below in figures 1-3.

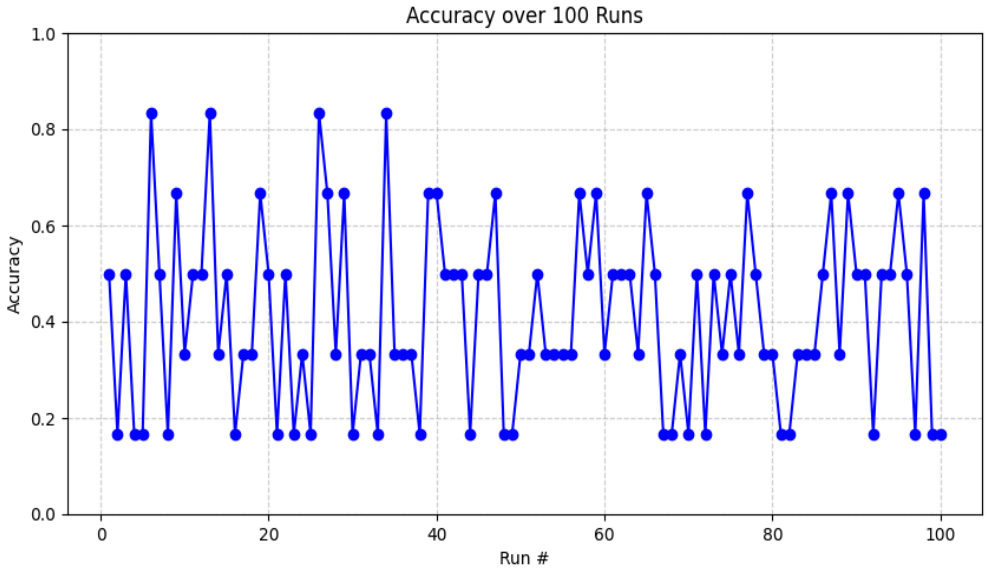


FIGURE 1
Classification accuracy over 100 runs using only the linguistic features (MLU, MLT ratio, pronoun%, conjunction%, negation%, total words). Each point represents the accuracy on one random 80/20 train-test split. The model's performance varies widely across splits, ranging from 16.7% up to 83.3%. The mean accuracy was 0.412 (41.2%), which is below chance level. This indicates that the neural network trained on these six linguistic features did not generalize well: in most trials it failed to do better than random guessing, highlighting the difficulty of the classification task with such limited data.

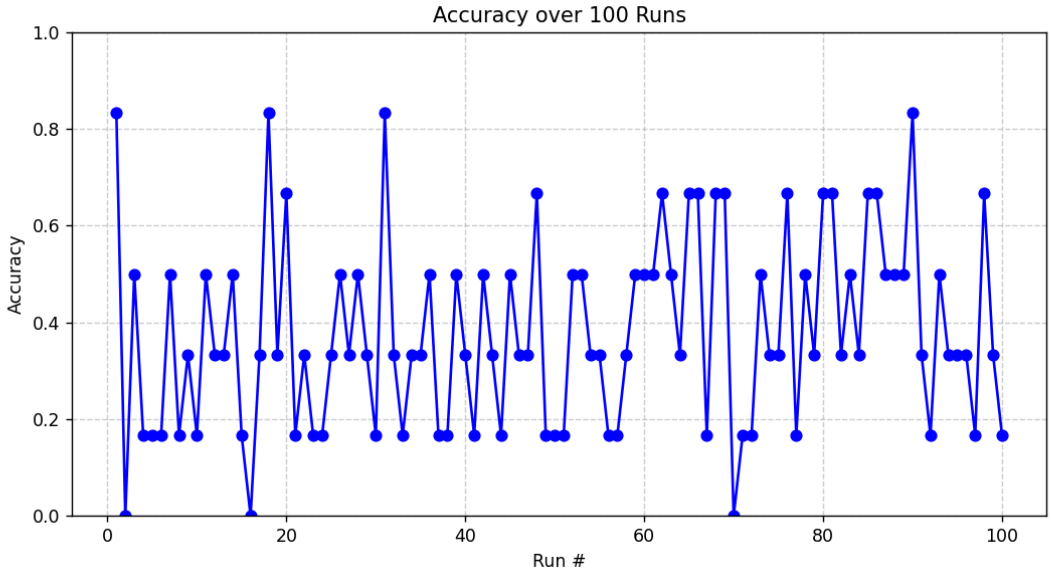


FIGURE 2

Classification accuracy over 100 runs using only the Zipfian features (α exponent, goodness-of-fit p-value, LR statistic, LR significance, and x_min). The model using solely the Zipf's law-based metrics also performed very unreliably. Accuracy across splits ranged from 0% (no correct predictions in a test set) up to ~83.3%, with a mean of about 0.380 (38.0%). This is again below the 50% chance baseline and shows extremely high variance. In many random splits, the Zipf-feature model completely failed to distinguish ASD from TD (yielding 0% accuracy), and in a few splits it achieved high accuracy seemingly by chance. Such instability underscores that the Zipf features alone did not provide a consistent discriminative signal for ASD in our sample.

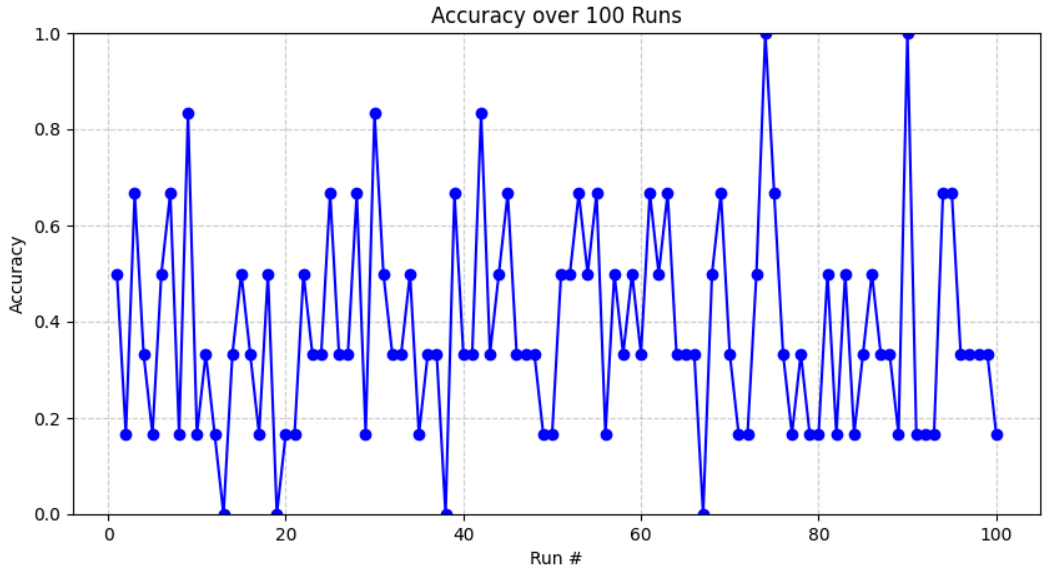


FIGURE 3

Classification accuracy over 100 runs using the combined feature set (all linguistic + Zipfian features). We had hypothesized that integrating both feature types might improve performance; however, the results did not bear this out. The combined model's accuracies ranged from 0.0 to 1.0 across the 100 trials, and the mean accuracy was approximately 0.385 (38.5%). Essentially, this model also hovered around chance performance. The fact that some trials reached 100% accuracy and others 0% accuracy suggests the model was overfitting in the training data of certain splits rather than learning a generalizable pattern. Overall, across all three feature conditions (linguistic-only, Zipf-only, and combined), the classifier failed to reliably distinguish ASD vs. TD from the transcript features, performing at or below what one would expect from random guessing.

In addition to classification accuracy, we analyzed the contribution of individual features and group differences to better understand these results. We conducted two-sample t-tests to compare the ASD and TD groups on each feature. Out of all the linguistic measures, only one showed a significant group difference: the usage of conjunctions was lower in the ASD group. Children with ASD used significantly fewer conjunctions per 100 words on average than typically developing children (ASD mean = 2.00, TD mean = 3.21; $t = -2.293$, $p = 0.0313$). This suggests that ASD children in our sample tended to produce simpler, less connected sentence structures (using fewer words like "and", "but", "because"). None of the other linguistic features exhibited statistically significant differences between the groups (all $p > 0.1$). For instance, the ASD group had slightly lower mean utterance lengths (MLU) and a marginally lower pronoun percentage than the TD group, but these differences were not reliable given the variability in the data. The mean length of turn ratio and the rate of negation words also did not differ significantly between ASD and TD. Likewise, the total number of words produced in the 100-utterance sample was highly variable and not significantly different between groups.

Similarly, none of the Zipfian distribution metrics differed significantly between ASD and TD groups (all $p > 0.14$ in two-sample comparisons). In other words, in this dataset, we did not observe a consistent deviation from Zipf's law in the

ASD children relative to typical children. Some ASD transcripts did show low Zipf fit p-values (indicating non-Zipfian word frequency distributions), but this was not uniform across the group – and notably, some TD children's transcripts also had poor Zipfian fits. The power-law exponent α of the word frequency distribution did not significantly differ between ASD and TD, nor did the likelihood ratio test outcomes on average. These statistical results align with the classification outcome that the Zipf features alone were insufficient: there was no clear, reliable separation in these metrics between the two classes.

To further interpret the model, we considered feature importance within the combined feature classifier. We found that the features related to Zipf's law were generally not dominant contributors in the neural network. When we inspected the trained models (for example, by observing weight magnitudes or performing importance tests across features), the Zipfian features tended to rank in the mid-to-lower tier in terms of influence on the prediction. In contrast, some traditional linguistic features (notably the conjunction usage rate) appeared among the more influential inputs for the model. This is consistent with the univariate analysis above. Since only conjunction rate showed a significant group difference, it makes sense that it would carry some predictive power, whereas the Zipf metrics, lacking clear group separation, did not substantially aid classification. Overall, the results indicate that neither set of features provided a strong, reliable signal for ASD, and combining them did not yield improvement – pointing to fundamental limitations in using these features (at least in isolation and with a small sample) for ASD detection.

IV. DISCUSSION

The findings of this study indicate that, contrary to our initial expectations and prior reports, simple linguistic features and Zipfian word-frequency features were not sufficient to reliably distinguish children with ASD from typically developing children in short speech transcripts. All of our classification models performed at or below chance level (~50% accuracy), a stark contrast to the ~75–80% accuracy reported by Ramesh and Assaf (2021) using similar transcript data. This failure to replicate suggests that there may be important differences in methodology or data that explain the discrepancy. One possibility is that the previous study's results were overly optimistic or specific to their particular dataset and feature set. Ramesh and Assaf used over 50 features and possibly leveraged a larger or more diverse set of transcripts (they included additional linguistic cues and might have combined multiple sources of child language data) [3]. In our replication, we deliberately focused on a smaller, controlled subset (16 ASD vs 16 TD children from the same corpus) and used a concise feature set drawn from their top predictors. It appears that with this small sample and limited feature set, the classification task is extremely challenging – to the point that even advanced modeling (a neural network with data augmentation) could not learn a generalizable distinction. This highlights a reproducibility issue: results from a single study, especially with small N, may not hold up when re-tested independently. Our outcome underscores the need for caution when interpreting impressive accuracy numbers from limited data; models can easily overfit or pick up spurious patterns that don't generalize.

Several factors likely contributed to the poor performance and lack of significant findings. Sample size is a primary limitation – with only 32 children (and 29 after filtering) and 100 utterances each, the statistical power to detect true differences is low, and machine learning models have very few examples to learn from. The high variance we observed (especially in the Zipf-only model's accuracy, ranging from 0% to 83% across trials; Figure 2) reflects the instability of models trained on such small data: results can fluctuate widely depending on which specific children are in the training vs. test set. In some splits the Zipf-based model happened to get lucky (for example, if one or two ASD children had very atypical word distributions that the model latched onto, it might score well on that split by chance), while in others it failed completely. This variability itself is a sign that the model was not truly learning a stable, generalizable pattern, but rather was sensitive to idiosyncrasies of the splits. In contrast, a robust classifier would show more consistent performance across different subsets of data. The linguistic-feature model was somewhat more stable but still hovered around chance (Figure 1), which suggests that the distinguishing power of those features (MLU, pronoun rates, etc.) was modest and easily swamped by noise given the small sample.

Another important factor is the limited length and context of the transcripts. We used only the first 100 utterances of each child's play session transcript. While this was done for consistency and to mirror prior work, 100 utterances (often short, simple child utterances) may not provide enough language sample to capture the full extent of a child's linguistic abilities or atypical patterns. Many language metrics, including Zipf's law characteristics, become more reliable with larger text samples. Zipf's law specifically is an asymptotic law – it describes the distribution of word frequencies in large corpora. In a very short transcript, the word frequency distribution is coarse and may not faithfully exhibit a clear Zipfian pattern. Our finding that many individual transcripts did pass Zipf's goodness-of-fit tests (appeared Zipfian) suggests that with limited data, even an ASD child's word usage can look "Zipf-like." De Palma et al. (2022) had reported that ASD speech deviated from Zipf's law [8], but they may have aggregated words across larger samples or multiple recordings to detect that deviation. In our child-level analysis, any subtle deviation was likely drowned out by variability. In essence, Zipfian features may hold theoretical promise, but they did not materialize as useful in our practical classification scenario, possibly because short transcripts do not provide stable estimates of the power-law parameters. Future studies could explore using longer transcripts or concatenating multiple language samples per child to get a more robust picture of their lexical distribution.

The one significant result we obtained – lower conjunction usage in ASD children's speech – is consistent with the notion that children with ASD produce simpler, less complex sentences. Conjunctions (like "and", "because", "but") are used to connect clauses and form more complex sentence structures. A lower rate of conjunctions in ASD transcripts suggests that these children might be using more short, simple sentences or not elaborating as much in conversation. This finding aligns with prior observations that ASD language can be less syntactically complex. For instance, Ramesh and Assaf's feature analysis noted differences in certain parts of speech usage [3], and our result specifically pinpoints conjunctions as significantly lower in ASD. Interestingly, we did not find a significant difference in pronoun usage, despite some reports of pronoun peculiarities in ASD (ex. pronoun reversal or avoidance). It's possible that with such young children (ages 3–6) and brief samples, pronoun usage is quite variable and hard to measure reliably, or that any ASD-specific pronoun patterns are subtle in this dataset. Similarly, Mean Length of Utterance (MLU) – while slightly lower for ASD on average – was not significantly different here, even though many studies identify lower MLU as a hallmark of ASD language delays. Again, this could be due to the small sample or the fact that the ASD children in this corpus might have relatively mild language

impairment (indeed, the ASD group's MLU had high variance, with some children near typical ranges). Overall, aside from conjunction usage, the linguistic differences between the ASD and TD transcripts in this small set appear subtle and inconsistent, which helps explain why the machine learning models struggled to separate the classes. In essence, the signal-to-noise ratio in this dataset is low – the true language differences exist (as evidenced by conjunction use) but are small compared to individual variability.

Our modeling strategy attempted to overcome the data scarcity through MCMC-based synthetic data augmentation and by using a neural network classifier. In hindsight, however, this approach also had limitations. Generating synthetic feature vectors from a Bayesian model assumes we can capture the feature distribution well, but with so few real samples, the estimated feature distribution (even if modeled with Bayesian uncertainty) may not reflect the true underlying population. The neural network, even with regularization, might then overfit to patterns in the augmented data that don't generalize. It is telling that a simpler model (like the logistic regression used in prior work [3]) reportedly achieved better results on what was likely the same or similar dataset. Complex models are not always the right choice for small data – in fact, they can memorize quirks rather than learn meaningful distinctions. Our neural network had many more parameters than the number of training examples, so it's quite possible that it effectively learned random variations or the majority class bias rather than any real linguistic pattern. Indeed, the combined model's accuracy being stuck around ~38–40% (Figure 3) suggests it might have been essentially guessing one class more often than the other (if the dataset or splits had slight class imbalances or idiosyncrasies). Without stronger features, the network had no reliable signal to latch onto, and sometimes its weight optimization may even have latched onto noise that led it to perform worse than chance.

When considering why our results diverged from earlier studies, a few key limitations of our study become clear. First, as mentioned, is the small sample size and limited data per child. Many machine learning applications in autism research that show high accuracy use either larger samples or additional data modalities [11]. Second, our feature set was restricted to a handful of linguistic measures (six core features, plus four Zipf metrics). It's possible that the true differences in ASD language are more complex and multi-dimensional than what these features capture. For example, pragmatic aspects of language (such as the appropriateness of responses, ability to maintain a topic, or turn-taking behavior beyond just length) and prosodic features (tone of voice, rhythm, intonation) are not captured in our transcript-based features. These could be important in distinguishing ASD speech but would require different analysis methods (ex. acoustic analysis for prosody, or detailed coding for pragmatic language use). Third, the homogeneity of the context (all children in a similar structured play session) might limit variability on some features. For instance, the total words spoken might depend heavily on how talkative the adult partner was or how the play session was facilitated, introducing noise. A more diverse set of conversational contexts or interactive settings might reveal differences not seen in a single-context study.

It is also worth noting that ASD is a spectrum with high heterogeneity. Within our 16 ASD children, there was likely a wide range of language abilities.Some children might have near-typical language, while others have severe delays or atypical speech patterns. With such a small sample, this heterogeneity makes it difficult for a classifier to find a one-size-fits-all pattern. One child with ASD might use very repetitive speech (skewing Zipf metrics), another might have relatively rich vocabulary but pragmatic oddities, and yet another might be largely non-verbal (producing very few utterances). Grouping them together and trying to classify against TD assumes a common signal, which may be too diffuse in a small sample. Larger datasets would enable stratifying ASD children by subgroups or developmental level, or using more personalized baselines.

How do these results inform the broader goal of automated ASD screening using speech?

On one hand, our negative findings temper the optimism from earlier studies that suggested a quick analysis of speech transcripts could accurately flag ASD. It appears that with just basic linguistic features from short transcripts, we cannot achieve reliable screening performance. This is important because it highlights that real-world applicability is not there yet – if one naively built a screening tool based on such features, it would likely have unacceptably high false positives and false negatives. On the other hand, our study is a reminder that reproducibility and validation on larger, independent samples are crucial before any computational tool is adopted in practice. The idea of using speech as a non-intrusive, privacy-preserving marker for ASD is still compelling (as we argued in the Introduction), but our work suggests that richer approaches are needed to realize it. It may not be as simple as feeding transcripts into a classifier; rather, one might need to combine multiple modalities or use more sophisticated language features to capture the complex differences in ASD communication.

Another direction for improvement is the use of advanced NLP (Natural Language Processing) techniques to derive features. Our feature set was relatively rudimentary (simple counts, ratios, and distribution fit measures). Modern NLP methods could

analyze semantic content, narrative structure, or leverage language models to detect atypical language usage patterns. For instance, one could use word embeddings or large language models to see if ASD transcripts differ in semantic coherence or topic distribution compared to TD transcripts. Those are beyond the scope of our current work but represent possible ways to capture differences that simple surface metrics miss. However, even advanced techniques will require sufficient training data to avoid overfitting and to learn the nuances of ASD language.

In summary, our discussion points to a few key implications: (a) The results call into question some of the earlier optimistic findings on ASD speech classification, showing that they may not generalize to all small-sample settings. (b) Feature sets that seemed promising (like basic linguistic measures or Zipf's law deviations) might not actually provide value for individual-level classification, especially with short samples. (c) To build a reliable ASD screening tool based on speech, we likely need larger datasets and a more comprehensive set of features (possibly multimodal, combining language with other behavioral cues). (d) Any proposed method must be rigorously validated on external samples to ensure it's capturing a true, generalizable signal of ASD and not just artifacts of a particular dataset. While the present study had negative findings, it offers a valuable reality check and guidance for future research: we should avoid "shortcuts" (like over-relying on one type of feature or a small dataset) and instead pursue a more holistic and well-validated approach. Despite the disappointing performance of the models in this study, we remain hopeful that with more data and more sophisticated analyses, automated speech-based tools could eventually contribute to early ASD identification. Our work illustrates the careful and thorough work still required to reach that goal.

V. CONCLUSION

Early identification of autism from natural speech remains a challenging goal. In this study, we attempted to replicate and extend previous work on automated ASD detection from child speech transcripts by incorporating both traditional linguistic features and novel Zipf's law-based lexical distribution features. Our results did not confirm the high accuracies reported by prior research – instead, all of our models performed around chance level in distinguishing 3–6 year-old children with ASD from typically developing peers using transcripts of 100 utterances each. The addition of Zipfian features, which we hypothesized would capture differences in word usage distributions, did not improve performance. In fact, these features showed no significant differences between ASD and TD groups in our sample and were not prioritized by the classifier. The only statistically significant linguistic marker we found was a lower usage of conjunctions in ASD speech, indicating simpler syntactic structure; however, this single feature was not enough to drive successful classification.

In conclusion, our findings suggest that the set of linguistic and Zipfian features examined – at least as operationalized on short transcripts – is insufficient for reliable ASD screening on its own. This small-scale study highlights the limits of reproducibility from earlier results and underscores the necessity of larger datasets and more nuanced feature extraction for this task. It may be necessary to combine multiple data sources (e.g., longer language samples, acoustic/prosodic features, or other behavioral cues) to achieve a robust automated screening tool for ASD. While speech remains a highly promising medium for non-intrusive and privacy-preserving ASD detection, simple transcript analysis alone did not yield a dependable classifier in our experiment. We recommend that future research focus on expanding the data and feature scope. For example, utilizing advanced language modeling techniques and exploring multimodal approaches, and ensuring thorough validation on independent cohorts. With more comprehensive efforts, it is hoped that accurate and objective early-screening methods for ASD can be developed. Our study serves as a reminder of the challenges involved and the careful work still required to reach that goal.

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